



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

This project analyzes SpaceX Falcon 9 launch data to predict first-stage landing success using Machine Learning. The goal is to support decision-making for space agencies evaluating competition with SpaceX.

Summary of methodologies

- Data collection via API & web scraping
- Data cleaning & transformation
- EDA with SQL & visualizations
- Folium map for launch site proximity
- Interactive dashboard with Plotly Dash
- ML models to predict landing outcome

Outputs:

- Analytical insights
- Visual summaries & dashboards
- Predictive model performance

Introduction

Project background and context

- The space industry is becoming increasingly competitive with the rise of **private space travel**. However, **launch cost** remains a major barrier for new entrants.
- **SpaceX** revolutionized this by enabling **reusable first-stage boosters**, reducing launch costs to **~\$62M**, compared to **\$165M+** for other providers — a key strategic advantage.

Problems you want to find answers

- Can we predict if Falcon 9's first stage will land successfully?
- How do variables like launch site, payload mass, or booster version impact landing success?
- Are there correlations between launch sites and success rates?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - SpaceX API
 - Web scrap Falcon 9 and Falcon Heavy launch records from Wikipedia
- Perform data wrangling
 - Converted mission outcomes into binary labels for model training (0 = fail, 1 = success)
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
- Prepared data (standardization, train/test split); created 'class' column; tested classifiers (LogReg, SVM, DT, KNN) to find best model

Data Collection

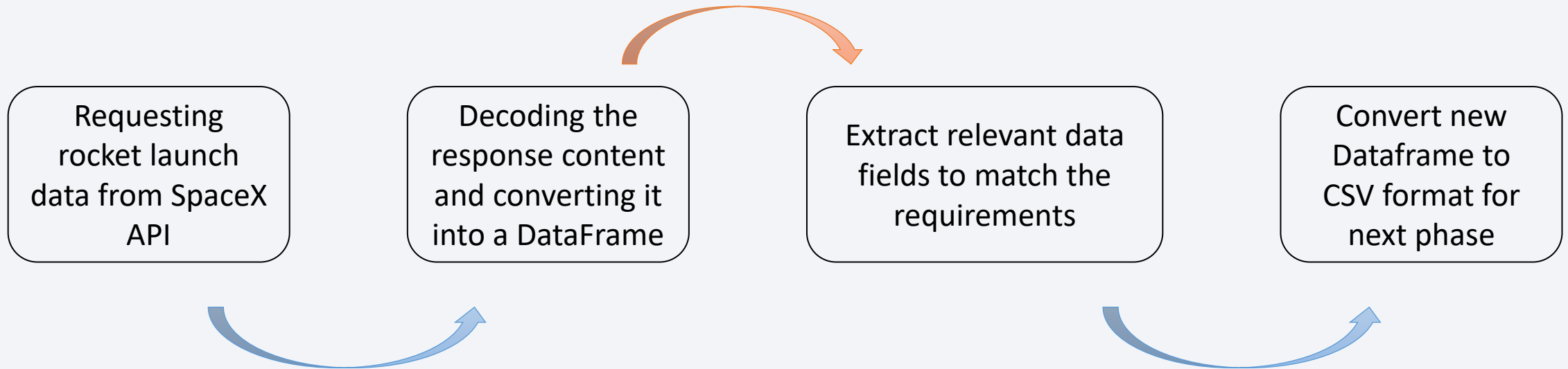
To develop the predictive model, we collected Falcon 9 launch data from two main sources:

- **SpaceX API:** Provided structured JSON data on past launches. The data was parsed, filtered for Falcon 9 missions, cleaned, and exported to CSV.
- **Wikipedia (Web Scraping):** Used BeautifulSoup to extract additional launch details from HTML tables, then transformed and saved the data in tabular format.

Both datasets were later merged and prepared for analysis and modeling.

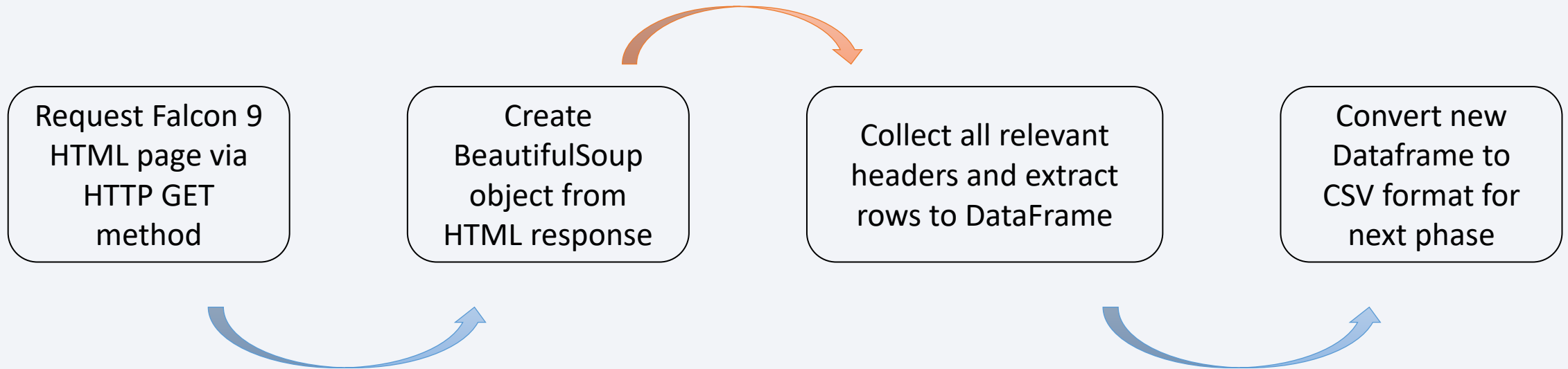
Data Collection – SpaceX API

[Data Collection API.ipynb](#)



Data Collection - Scraping

Data Collection Webscraping.ipynb



Performed Exploratory Data Analysis (EDA) to identify patterns and prepare the dataset for modeling.

Converted mission outcomes into binary training labels for supervised learning:

- 1 → Successful landing
- 0 → Unsuccessful landing

Landing Scenarios Mapped to Labels:

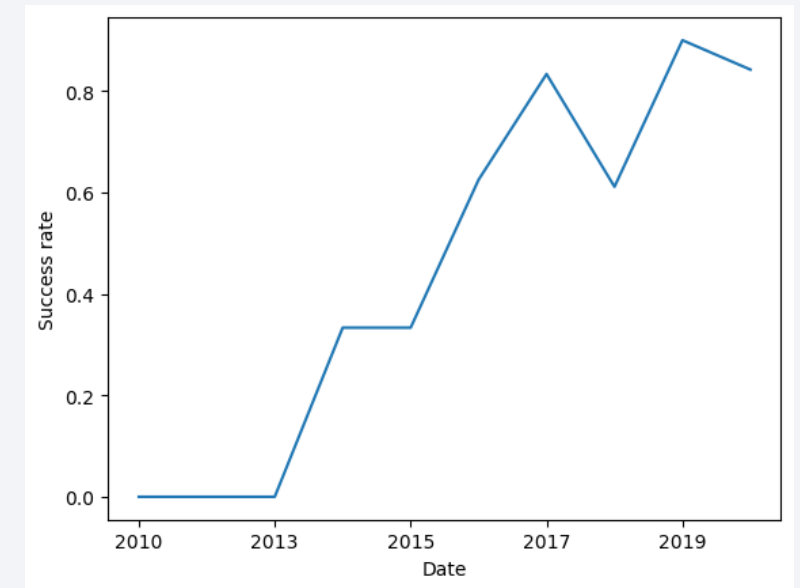
- ✓ True Ocean → Successful ocean landing
- ✗ False Ocean → Failed ocean landing
- ✓ RTLS (Return to Launch Site) → Successful ground pad landing
- ✗ False RTLS → Failed ground pad landing
- ✓ True ASDS → Successful drone ship landing
- ✗ False ASDS → Failed drone ship landing

This transformation enabled consistent labeling of outcomes for classification models.

EDA with Data Visualization

[EDA Data Visualization.ipynb](#)

- To better understand the data, several visualizations were created:
- **Scatter Plots:**
 - Explored relationships between variables such as:
 - *Flight Number vs Launch Site*
 - *Payload vs Launch Site*
 - *Flight Number vs Orbit*
 - *Payload vs Orbit*
- **Bar Chart:**
 - Compared **success rates by orbit type**.
- **Line Chart:**
 - Analyzed the **trend of average launch success over time**.
- These visuals helped uncover patterns and informed feature selection for modeling.



To explore and summarize the dataset, several SQL queries were executed on an **IBM DB2 cloud instance**, including:

- Retrieval of **unique launch sites** and filtering records by site prefix (e.g., 'CCA').
- Aggregation of **payload mass** by specific missions and booster versions (e.g., NASA CRS, F9 v1.1).
- Identification of:
 - Date of the **first successful ground pad landing**
 - **Boosters with successful drone ship landings** and specific payload range
 - **Total success and failure counts** in mission outcomes
- Advanced queries:
 - Subquery to find **booster version with max payload**
 - Filter **failed drone ship landings in 2015**
 - **Rank landing outcomes** within a time window (2010–2017)

These operations helped uncover key metrics and support feature engineering for model training.

Build an Interactive Map with Folium

[Launch Site Location.ipynb](#)

An interactive map was built using Folium to explore how the location and proximity of launch sites relate to launch success.

Key Features Implemented:

- Marked all launch sites on the map using Marker and Circle.
- Used MarkerCluster() to show success (green) and failure (red) outcomes.
- Calculated distances from launch sites to nearby coastlines, railroads, highways, and cities.
- Added mouse coordinates and distance markers using MousePosition() and Marker().
- Used Polyline() to draw lines between launch sites and surrounding infrastructure.

Insights Gained:

- ✓ Launch sites are close to coastlines, railways, and highways
- ✓ Launch sites maintain safe distance from cities

Build a Dashboard with Plotly Dash

[spacex-dash-app.py](#)

- A **Plotly Dash web app** was developed to explore SpaceX launch data interactively and in real time.
- **Main Components:**
- **Launch Site Dropdown:** Filters visuals by site or shows all data.
- **Pie Chart:** Shows total successes (all sites) or success vs failure (per site).
- **Payload Range Slider:** Filters by payload to reveal success trends.
- **Scatter Plot:** Displays relationship between payload and mission outcome, colored by booster version.
- **Key Insights from Dashboard:**
- 🚀 Most successful launches: **KSC LC-39A** (10 launches)
- 📊 Highest success rate: **KSC LC-39A** (76.9%)
- ✅ Highest success by payload: **2000–5000 kg**
- ❌ Lowest success: **0–2000** and **5500–7000 kg**
- 🏆 Best booster version: **FT**

Predictive Analysis (Classification)

[SpaceX Machine Learning.ipynb](#)

Read dataset and
create a 'Class'
array

Standardize the
data

Train/Test Split
data

Create and Refine
Models

Find the best
model

```
data.head()
```

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40

```
Y = data['Class'].to_numpy()
```

```
# students get this
transform = preprocessing.StandardScaler()
X = transform.fit_transform(X)
```

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.2, random_state = 2)
```

```
# Crear un diccionario con los resultados
results = {
    'Modelo': ['Logistic Regression', 'SVM', 'Decision Tree', 'KNN'],
    'Mejor Score (CV)': [log_score, svm_score, tree_score, knn_score],
    'Score en Test': [log_test_score, svm_test_score, tree_test_score, knn_test_score]
}
```

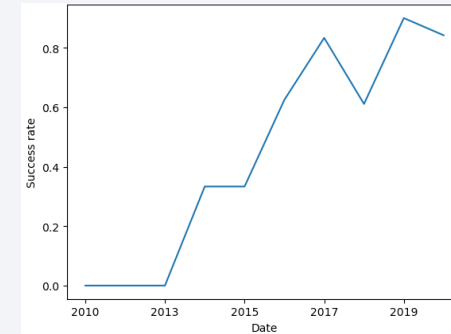
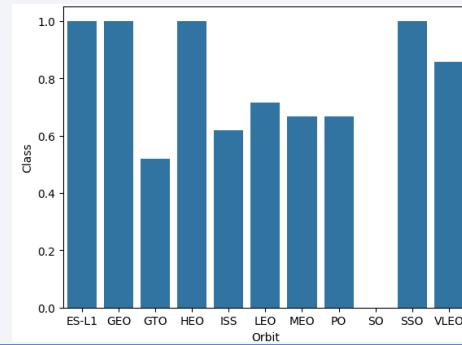
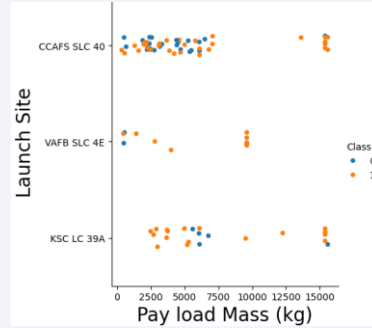
```
# Crear DataFrame
df_resultados = pd.DataFrame(results)
```

```
# Mostrar la tabla
print(df_resultados)
```

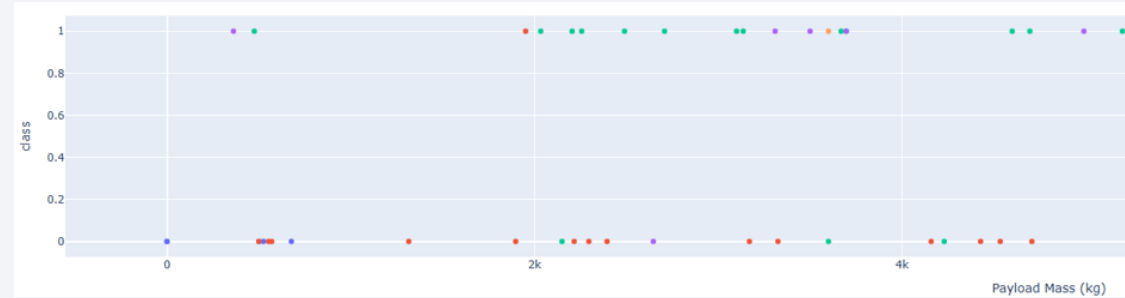
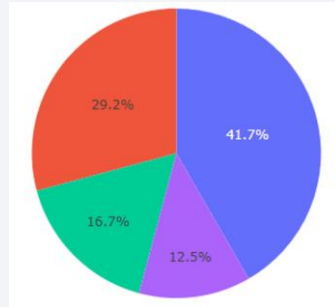
	Modelo	Mejor Score (CV)	Score en Test
0	Logistic Regression	0.846429	0.833333
1	SVM	0.848214	0.833333
2	Decision Tree	0.889286	0.833333
3	KNN	0.848214	0.833333

Results

Exploratory data analysis results

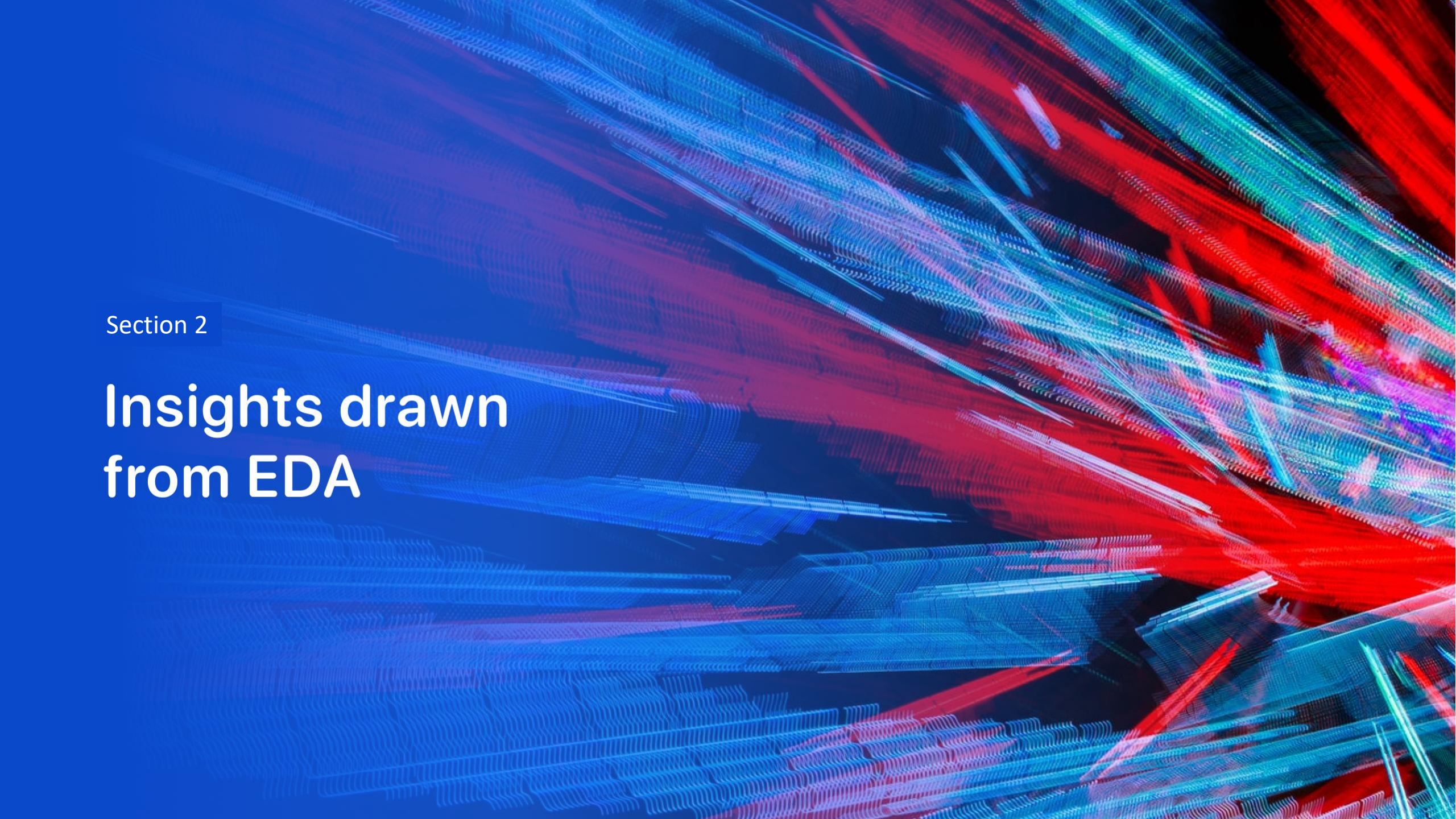


Interactive analytics demo



Predictive analysis results

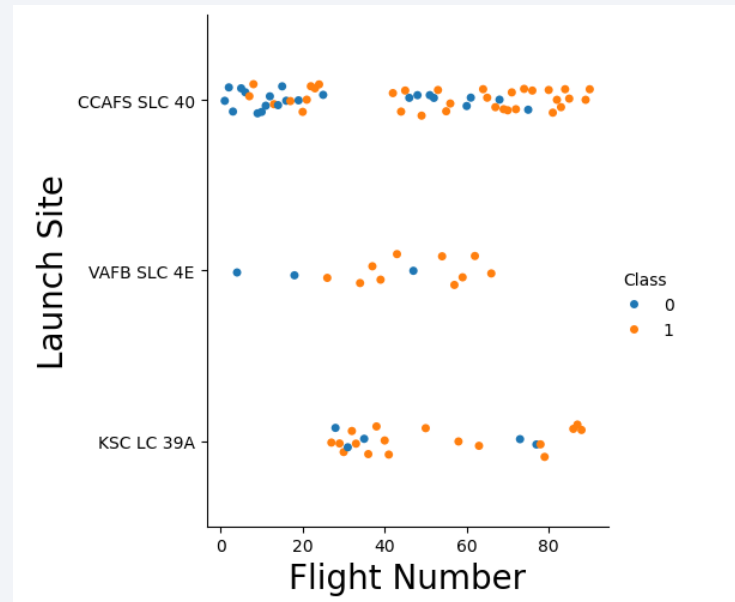
	Model	Best Score (CV)	Test Score
0	Logistic Regression	0.846429	0.833333
1	SVM	0.848214	0.833333
2	Decision Tree	0.889286	0.833333
3	KNN	0.848214	0.833333

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

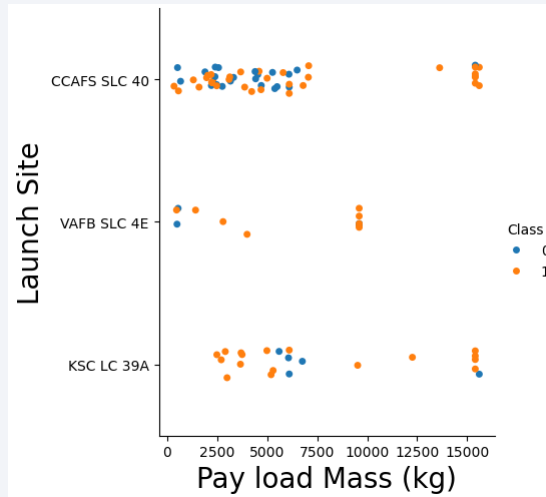


This plot shows the relationship between Flight Number and Launch Site, with color-coded markers representing landing success (1) or failure (0).

- Each dot is a launch; orange indicates success, blue indicates failure.
- KSC LC 39A shows mostly successful outcomes in later flights.
- CCAFS SLC 40 has a high number of both successes and failures spread across flight numbers.
- VAFB SLC 4E shows fewer launches with mixed outcomes.

This visualization helps identify trends in launch performance over time and by site.

Payload vs. Launch Site



This chart shows the distribution of **Payload Mass (kg)** by **Launch Site**, with outcome classes:

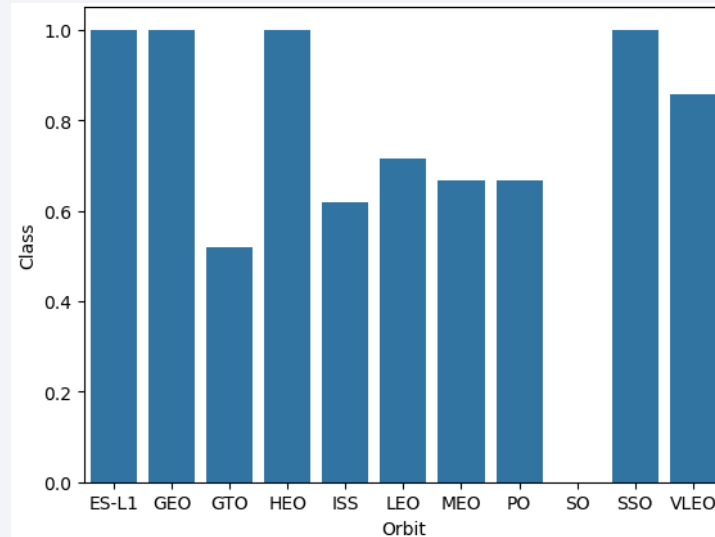
- **Orange = Success (1)**
- **Blue = Failure (0)**

Observations:

- Most payloads fall between **2000–8000 kg** across all sites.
- **CCAFS SLC 40** shows the highest variability in payload mass, including the largest payloads.
- **KSC LC 39A** and **VAFB SLC 4E** operate with more moderate payload ranges.
- Successful landings (orange) are frequent in the mid-range payloads across all sites.

This helps analyze how payload weight may impact landing outcomes at different launch locations.

Success Rate vs. Orbit Type



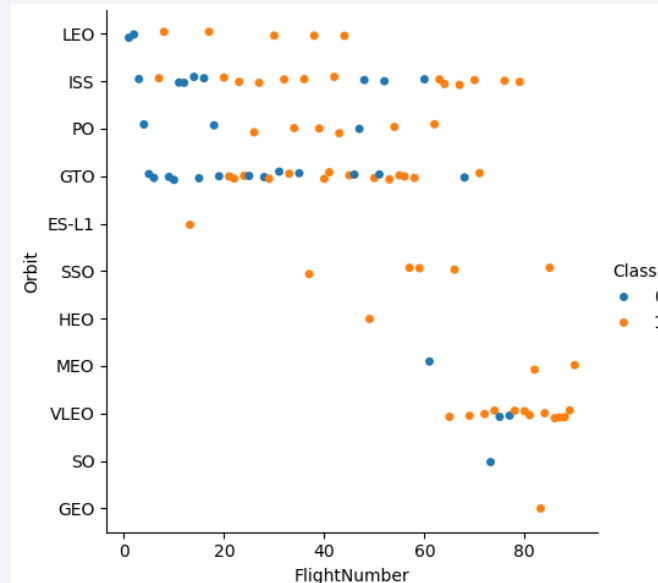
This chart displays the **average mission success rate (Class)** for each **Orbit Type**.

Key Insights:

- Orbits like **ES-L1**, **GEO**, **HEO**, and **SSO** achieved a **100% success rate**.
- **GTO (Geostationary Transfer Orbit)** has the **lowest success rate**, under 0.6.
- Other orbits such as **LEO**, **ISS**, and **VLEO** show moderate to high success rates.

This visualization helps compare mission reliability across different orbital destinations.

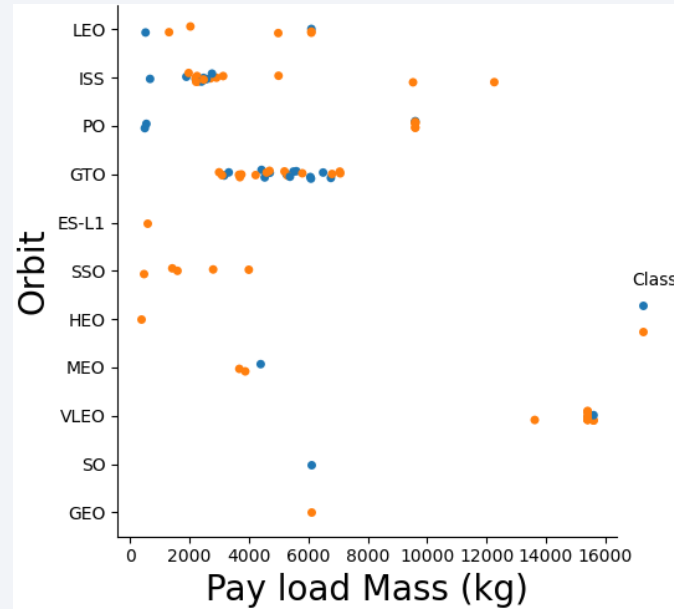
Flight Number vs. Orbit Type



This scatter plot shows **launch outcomes** across **orbit types** over time.

- Early flights had **more failures**, especially in **LEO**, **ISS**, and **GTO**.
- **Success rates improve** in later flights, notably for **VLEO**, **SSO**, and **MEO**.
- Less frequent orbits like **GEO** and **ES-L1** show mostly **successful launches**.

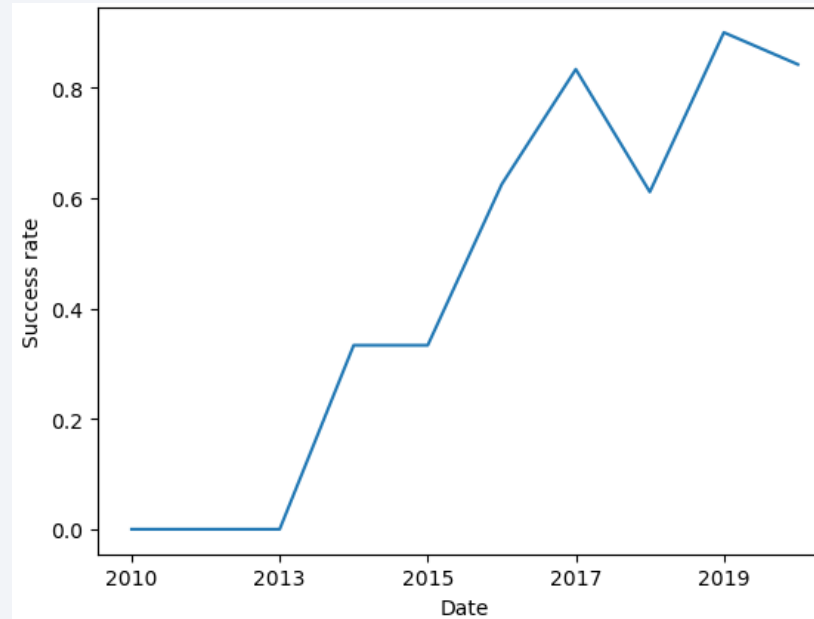
Payload vs. Orbit Type



This scatter plot shows how **payload mass** varies across **orbit types**, with success (●) and failure (●) outcomes.

- Most payloads range between **2000–8000 kg**, especially in **GTO**, **ISS**, and **PO**.
- **Higher payloads** (up to 16,000 kg) appear in **VLEO**, with mostly **successful outcomes**.
- No clear pattern links payload mass to failure—success occurs across all ranges.

Launch Success Yearly Trend



This line chart shows the **yearly trend** of SpaceX launch success rates.

- From **2010 to 2013**, the success rate remained at **0**.
- A sharp increase began in **2014**, reaching over **80% by 2017**.
- After a dip in **2018**, the success rate peaked near **90% in 2019**.

This trend reflects continuous **improvements in launch reliability** over time.

All Launch Site Names

- 'distinct' returns only unique values from the queries column (Launch_Site)
- There are 4 unique launch sites

Query: SELECT DISTINCT Launch_Site FROM SPACEXTBL

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Using keyword 'Like' and format 'CCA%', returns records where 'Launch_Site' column starts with "CCA".
- Limit 5, limits the number of returned records to 5.

Query: `SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5`

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- 'sum' adds column 'PAYLOAD_MASS_KG' and returns total payload mass for customers named 'NASA (CRS)'

Query: SELECT SUM(PAYLOAD_MASS_KG_) FROM SPACEXTBL WHERE Customer = 'NASA (CRS)'

SUM(PAYLOAD_MASS_KG_)

45596

Average Payload Mass by F9 v1.1

- 'avg' keyword returns the average of payload mass in 'PAYLOAD_MASS_KG' column where booster version is 'F9 v1.1'

Query: SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1%'

AVG(PAYLOAD_MASS_KG_)
2534.66666666666665

First Successful Ground Landing Date

- 'min(Date)' selects the first or the oldest date from the 'Date' column where first successful landing on group pad was achieved
- Where clause defines the criteria to return date for scenarios where 'Landing_Outcome' value is equal to 'Success (ground pad)'

MIN(DATE)

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- The query finds the booster version where payload mass is greater than 4000 but less than 6000 and the landing outcome is success in drone ship
- The 'and' operator in the where clause returns booster versions where both conditions in the where clause are true

Query: SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

This query **groups landing outcomes** into three categories: **Success**, **Failure**, or **Other**, and **counts the total number of records** in each category.

Query:

```
SELECT
CASE
  WHEN Landing_Outcome IN ('Success', 'Success (ground pad)', 'Success (drone ship)') THEN 'Success'
  WHEN Landing_Outcome IN ('Failure', 'Failure (parachute)', 'Failure (drone ship)', 'Uncontrolled (ocean)') THEN 'Failure'
  ELSE 'Other'
END AS Outcome_Category,
COUNT(*) AS Total
FROM SPACEXTBL
GROUP BY Outcome_Category;
```

Outcome_Category	Total
Failure	12
Other	28
Success	61

Boosters Carried Maximum Payload

- The sub query returns the maximum payload mass by using keyword 'max' on the pay load mass column
- The main query returns booster versions and respective payload mass where payload mass is maximum with value of 15600

Query: SELECT Booster_Version FROM SPACEXTBL WHERE
PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_)
FROM SPACEXTBL)

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

This query **filters launch records from 2015** where the **landing failed on a drone ship**, and returns the **month, landing outcome, booster version, and launch site** for each case.

Query: `SELECT substr(Date, 6,2) AS 'Mes', Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE substr(Date, 0, 5) = '2015' AND Landing_Outcome = 'Failure (drone ship)'`

Mes	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The 'group by' key word arranges data in column 'Landing__Outcome' into groups
- The 'between' and 'and' keywords return data that is between 2010-06-04 and 2017-03-20
- The 'order by' keyword arranges the counts column in descending order
- The result of the query is a ranked list of landing outcome counts per the specified date range

Query: SELECT Landing_Outcome, count(Landing_Outcome) FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY count(Landing_Outcome) DESC

Landing_Outcome	count(Landing_Outcome)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

SpaceX Falcon 9 Launch Sites Map



Figure 1 shows a global map with Falcon 9 launch sites in **California and Florida**, each marked with a **circle, label, and popup**. All sites are located **near the coast**.

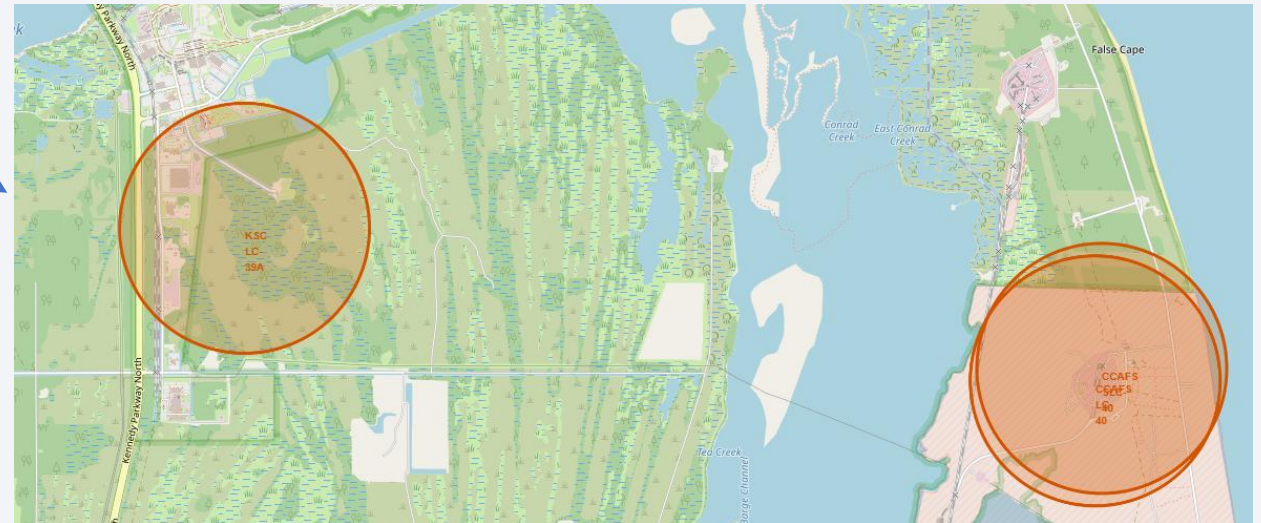
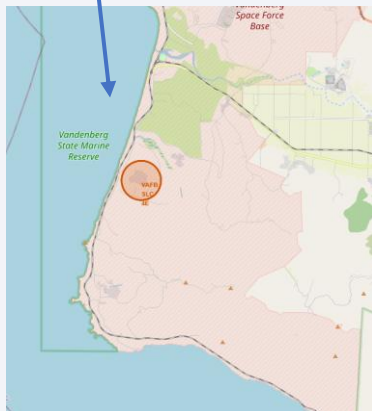
Figures 2 and 3 zoom in to display the 4 specific launch sites:

VAFB SLC-4E (CA)

CCAFS LC-40 (FL)

KSC LC-39A (FL)

CCAFS SLC-40 (FL)



SpaceX Falcon 9 Success/Failed Launch Map

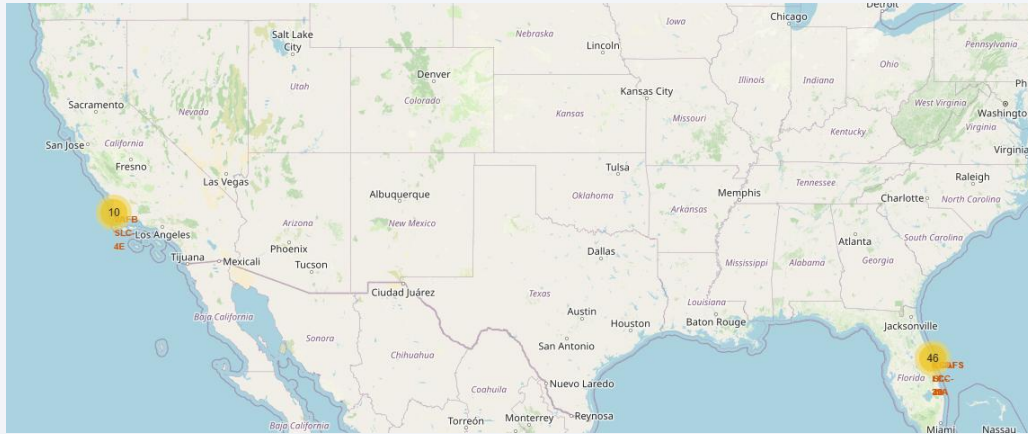
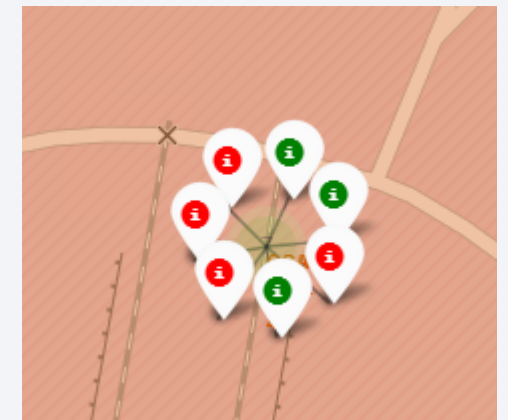
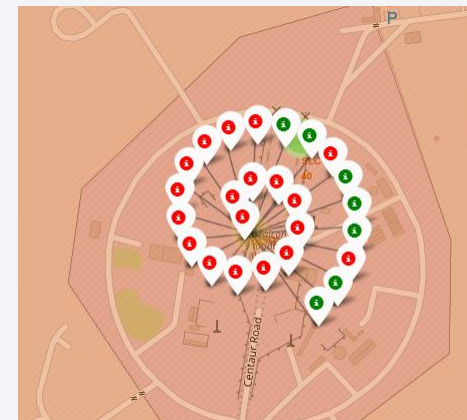
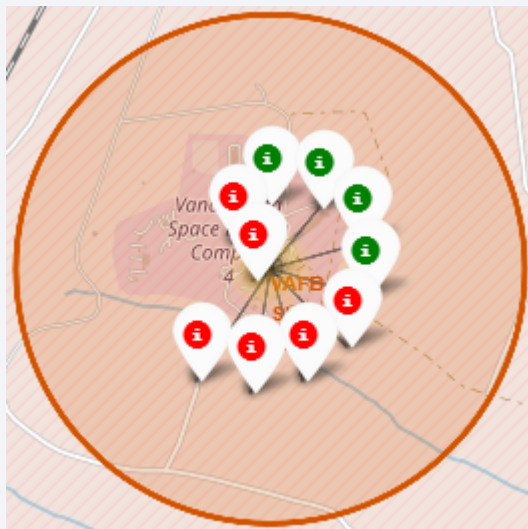
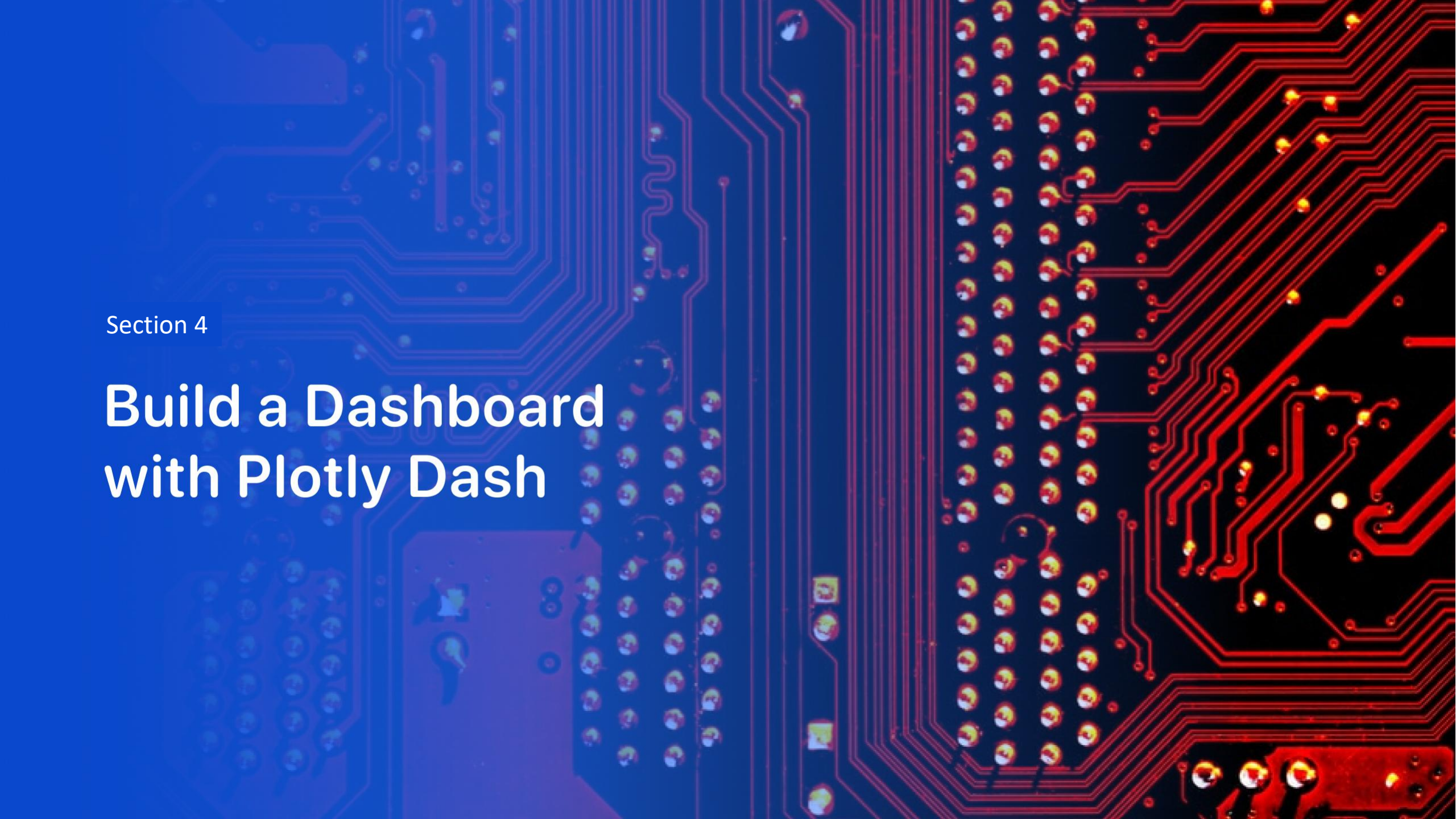


Figure 1 shows a **US map** with all launch sites, where numbers indicate **total successes and failures**. **Figures 2–5** zoom in on each site, using **green markers for success** and **red for failure**. **KSC LC-39A** stands out with the **highest number of successful launches**.

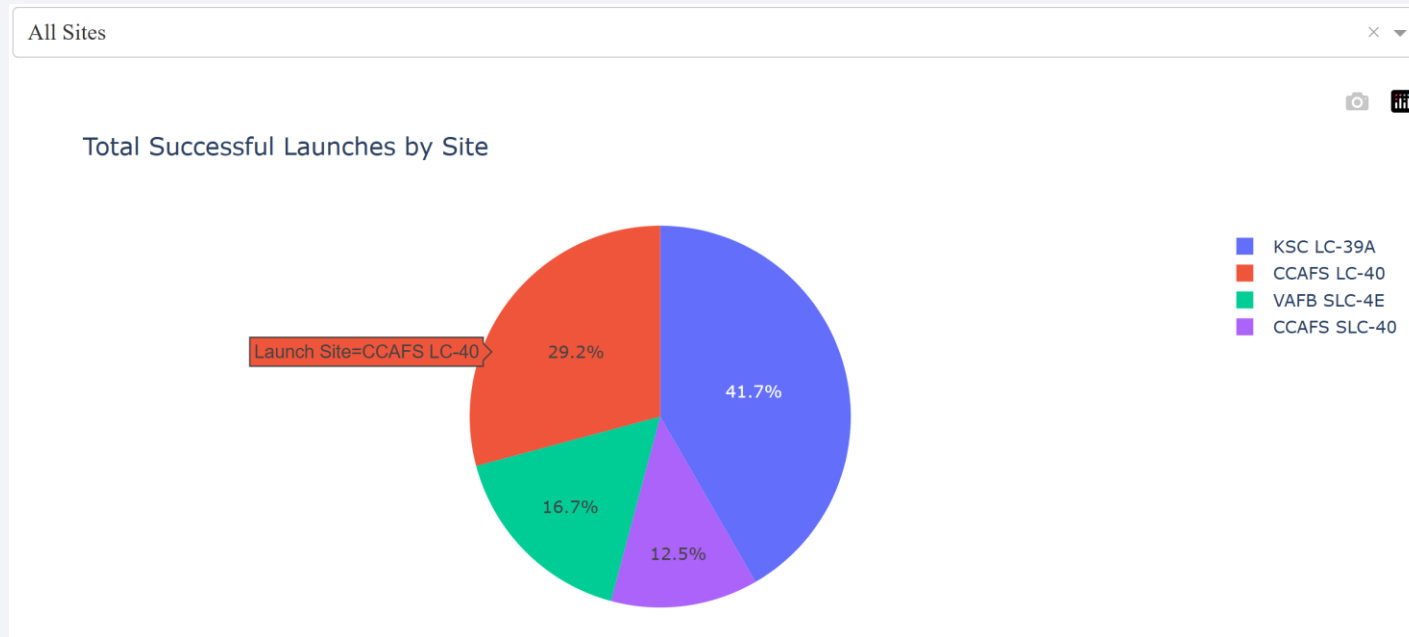




Section 4

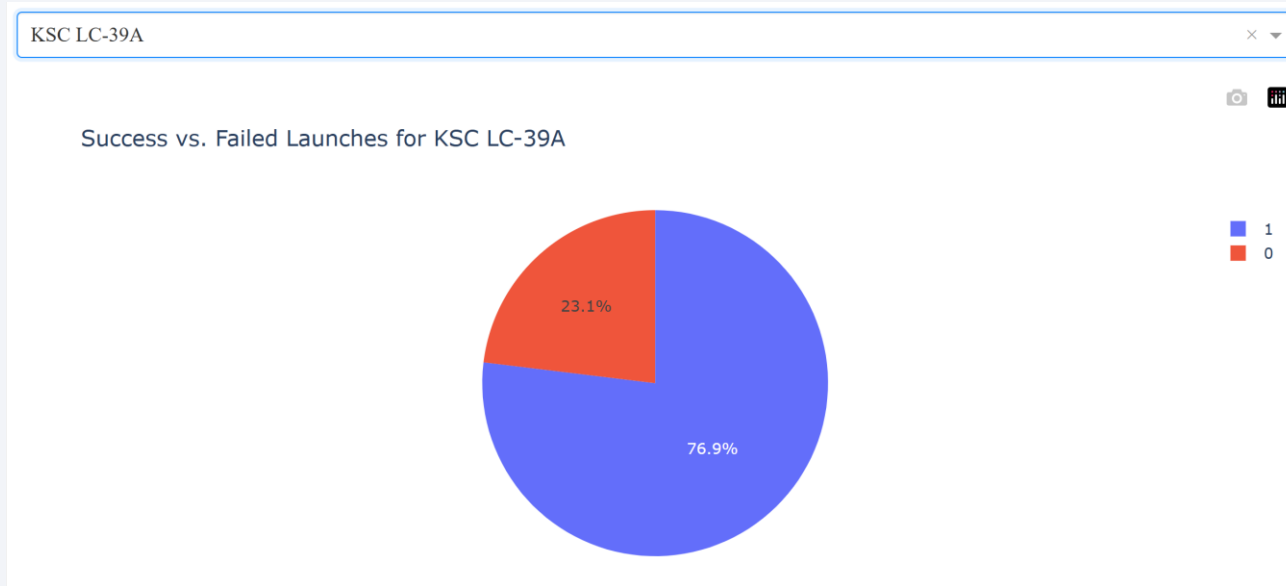
Build a Dashboard with Plotly Dash

Launch Success Counts



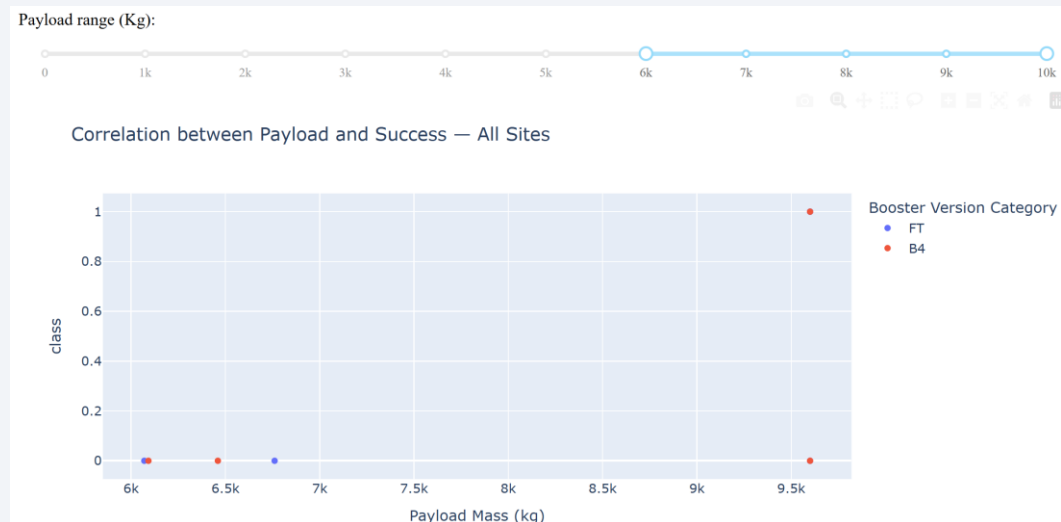
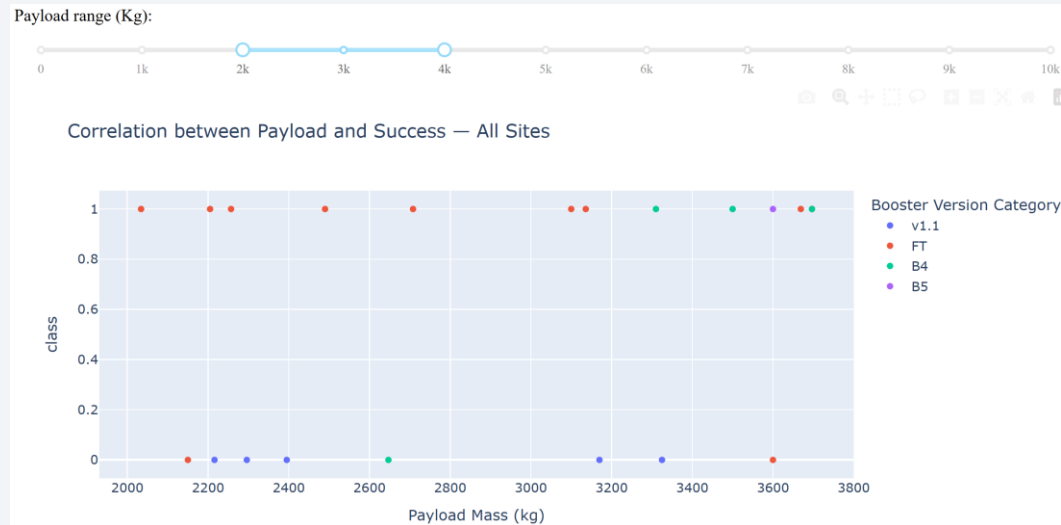
- Launch Site 'CCAFS SLC 40' has the lowest launch success rate.
- Launch Site 'KSC LC-39A' has the highest launch success rate

KSC-LC-39A Launch Success Ratio



- Launch success rate is 76.9%
- Launch success failure rate is 23.1%

Payload vs Launch Outcome



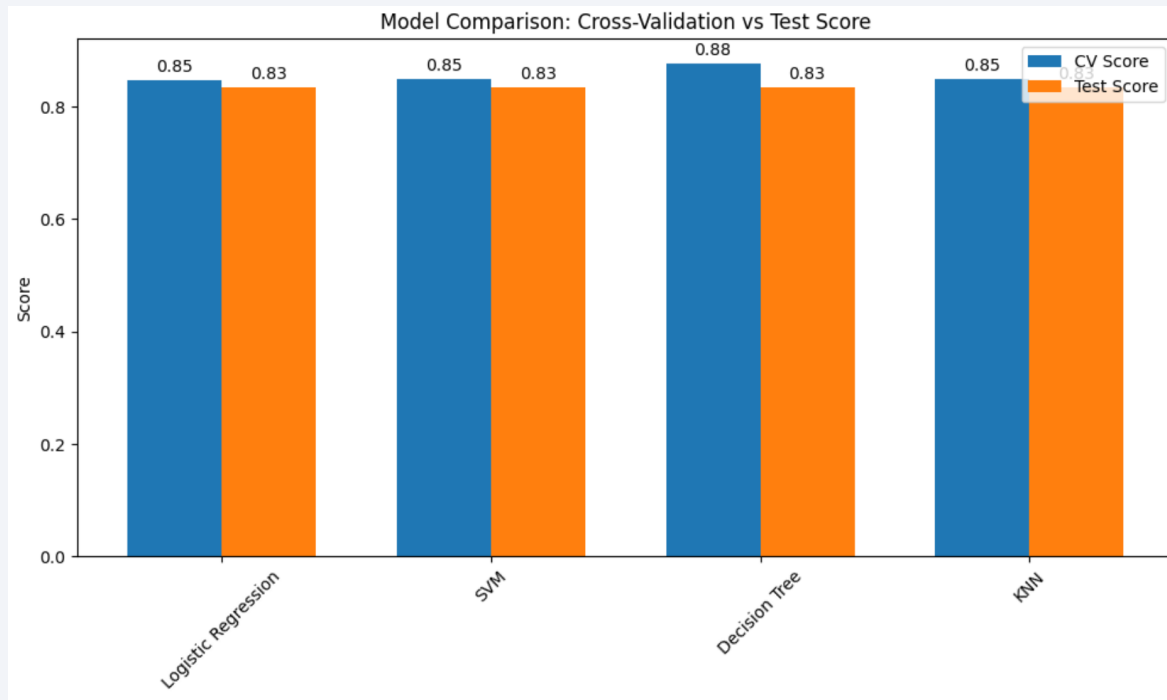
- In the first image (2000–3800 kg), most launches were successful, especially with the FT booster version.
- In the second image (6000–9500 kg), failures are more frequent, suggesting higher payloads may have lower success rates.
- Booster versions B4 and FT appear in both ranges but show more failures with heavier payloads.



Section 5

Predictive Analysis (Classification)

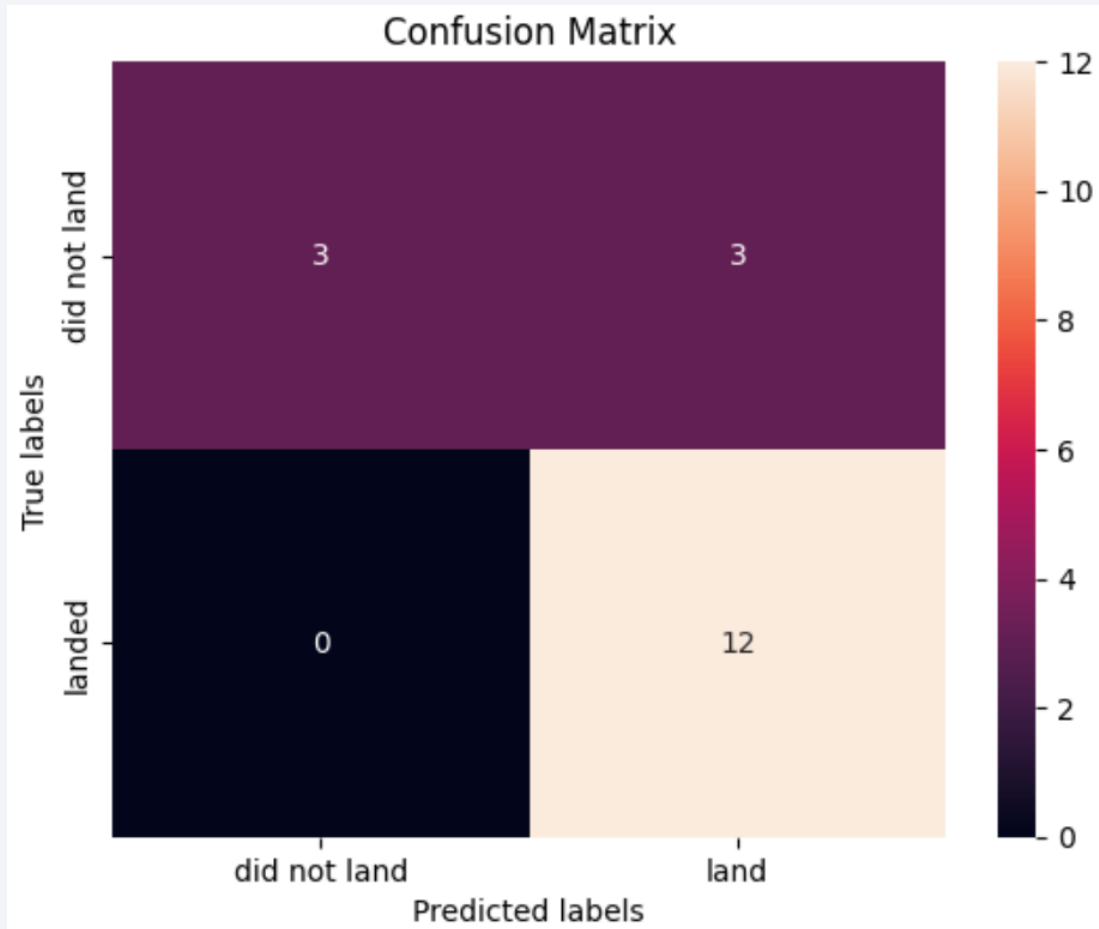
Classification Accuracy



- All models achieved the same test score of **0.833**, indicating similar generalization performance on unseen data. However, the Decision Tree had the highest cross-validation score (**0.876**), suggesting it performed best during training and validation. While this may point to a slightly better fit, the uniform test scores imply that no model significantly outperforms the others in real-world predictions.

	Model	Best Score (CV)	Test Score
0	Logistic Regression	0.846429	0.833333
1	SVM	0.848214	0.833333
2	Decision Tree	0.876786	0.833333
3	KNN	0.848214	0.833333

Confusion Matrix



- The model correctly predicted 12 landings and 3 non-landings, but misclassified 3 landings as failures. It shows high precision for successful landings, but some errors predicting failed landings.

Conclusions

- As the number of launches increases, the likelihood of a successful first-stage landing improves, suggesting operational maturity over time.
- Although higher payloads (especially in the 2000–5000 kg range) tend to have better success rates, the correlation between payload mass and landing success is not consistent across all weight ranges.
- The overall launch success rate improved dramatically from 2013 to 2020, reaching over 85% by 2019, indicating significant progress in SpaceX's reliability.
- Among all launch sites, KSC LC-39A recorded the highest success rate, while CCAFS SLC-40 showed more variability with a mix of successful and failed missions. Orbits such as ES-L1, GEO, HEO, and SSO consistently achieved perfect or near-perfect success, whereas GTO stood out with the lowest reliability.
- Launch sites are strategically positioned near coastlines, highways, and railroads, but are safely distanced from cities, likely to minimize risk and improve logistics.
- The Decision Tree classifier outperformed other models with a cross-validation score of ~87.6%, and all models achieved around 83% accuracy on test data. However, with a limited dataset, further improvements could be achieved through model tuning and additional data.

Thank you!

